Spin Hall - Littlewood Polynomials

1. Introduction

When studying Schur polynomials, we mentioned following identity, now referred to as a Cauchy-type identity.

$$\sum_{\substack{\lambda \\ \ell(\lambda)\leq N}} s_\lambda(u_1,\dots,u_N)\, s_\lambda(v_1,\dots,v_N) = \prod(u,v) = \prod_{i,j}(1-u_i v_j)^{-1}$$

Next, for the Hall-Littlewood polynomials

$P_\lambda(u;q) = \dfrac{1}{v_\lambda(q)} \displaystyle\sum_{\sigma\in S_n} \sigma\left(u^\lambda \prod_{i<j}\frac{u_i - q u_j}{u_i - u_j}\right)$, where $v_\lambda(q) = \displaystyle\prod_{i\geq 0} v_{m_i(\lambda)}(q)$ with

$m_i(\lambda)$ = # of parts of λ equal to i, $v_r(q) = \dfrac{(q;q)_r}{(1-t)^r}$ and $(q;q)_r = (1-q)(1-q^2)\cdots(1-q^r)$,

we also had a similar Cauchy-type identity:

$$\sum_{\substack{\lambda \\ \ell(\lambda)\leq N}} b_\lambda(q)\, P_\lambda(u;q)\, P_\lambda(v;q) = \prod_{i,j}\frac{1-q u_i v_j}{1-u_i v_j},$$

where $b_\lambda(q) = \displaystyle\prod_{i\geq 0}(q;q)_{m_i(\lambda)}$.

Finally, for Macdonald polynomials, we had

$$\sum_{\substack{\lambda \\ \ell(\lambda)\leq N}} b_\lambda(q)\, P_\lambda(u;t,q)\, P_\lambda(v;t,q) = \prod(u,v;t,q) = \prod_{i,j}\frac{(q u_i v_j;t)_\infty}{(u_i v_j;t)_\infty}$$

Our goal is to introduce a new type of symmetric function, called spin Hall-Littlewood symmetric rational function and show such functions satisfy a similar Cauchy type identity.

Def.: (Signatures) A signature of length N is a sequence of $(\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_N)$, $\lambda_i \in \mathbb{Z}$. Let Sign_N denote the set of nonnegative signatures, i.e., the ones where $\lambda_i \geqslant 0$ for all i. Let us define $\mathrm{Sign}_0 = \{\emptyset\}$.

We also define the notation $\lambda = (0^{m_0(\lambda)} 1^{m_1(\lambda)} 2^{m_2(\lambda)} \dots)$, with $m_i(\lambda) =$ # of parts in λ equal to i as before.

Define a spin Hall-Littlewood symmetric rational function by the formula

convention, $\lambda_i = 0 \Rightarrow$ product $= 1$

$$F_\lambda(u_1, u_2, \dots, u_N; q) = \sum_{\sigma \in S_N} \sigma\left(\prod_{i<j} \frac{u_i - q u_j}{u_i - u_j} \prod_{i=1}^{N} \left(\frac{1-q}{1 - s_{\lambda_i} \xi_{1+\lambda_i} u_i} \prod_{j=0}^{\lambda_i - 1} \frac{\xi_j u_i - s_j}{1 - \xi_j s_j u_i} \right) \right)$$

The action of $\sigma \in S_N$ permutes the variables u_i, but has no effect on ξ_i and s_i. The function is symmetric by definition and it is obtained as a deformation of the usual Hall-Littlewood symmetric functions. Each F_λ depends on the variables u_i, the quantum parameter q, the inhomogeneity parameters ξ_x and s_x, $x \in \mathbb{Z}_{\geq 0}$ where s_x is called the spin parameter. In general, ξ_x and s_x are thought to be arbitrary complex numbers, but for more interesting results, additional conditions will be imposed soon. Note that if $s_x = 0$ and $\xi_x = 1$ for all x, then F_λ becomes the usual Hall-Littlewood symmetric functions.

The motivation behind these functions comes from statistical mechanics, where they appear naturally as the partition functions of a specific type of integrable lattice model: higher spin six-vertex model. The definition of this model, as well as its version of the Young-Baxter Equation, will be vital to understand the Cauchy-type identity for the F_λ.

One can define a few important variations on spin Hall-Littlewood polynomials. For reasons that will be presented later, one can define the so called dual Hall-Littlewood symmetric functions in terms of the regular Hall-Littlewood functions as

$$F^{*}_{\lambda}(v_1,\dots,v_N) = \prod_{r\geq 0}\frac{(s_r^2;q)_{m_r}}{(q;q)_{m_r}}\cdot F_{\lambda}(v_1,\dots,v_N)\Big|_{\xi_x\to\xi_x^{-1}\text{ for all } x},$$

where $\lambda = (0^{m_0}1^{m_1}2^{m_2}\dots)$.

There are also the stable spin Hall-Littlewood functions, which are indexed by partitions, rather than signatures. The distinction is made as partitions only care about nonzero parts of a nondecreasing sequence of nonnegative integers, and partitions may have an arbitrary length. As a result, such stable functions may depend on an arbitrary number of variables $k\geq \ell(\lambda)$, not tied to the length of the partition. By convention, if $k<\ell(\lambda)$ the stable function is identically zero. We denote the stable functions by $\tilde{F}_{\lambda}$, $\tilde{F}^{*}_{\lambda}$.

We can define stable functions in terms of the regular functions in multiple ways. The most straightforward is (here, $\lambda=(\lambda_1\geq\lambda_2\geq\dots\geq\lambda_\ell>0$ is a partition).

$$\tilde{F}_{\lambda}(u_1,\dots,u_k) = \frac{1}{(q;q)_{k-\ell}}\cdot F_{\lambda\cup 0^{k-\ell}}(u_1,\dots,u_k)\Big|_{s_0=0}$$

$$= \sum_{\sigma\in S_k/S_\lambda}\sigma\left(\prod_{i<j}\frac{u_i-qu_j}{u_i-u_j}\prod_{i=\ell+1}^{k}\left(\frac{1-q}{1-s_0\xi_0 u_i}\right)\right) = \frac{(1-q)^{k-\ell}}{\prod_{i=\ell+1}^{N}(1-s_0\xi_0 u_i)}\cdot\left[\sum_{\sigma}\left(\prod_{i<j}\frac{u_i-qu_j}{u_i-u_j}\right)\right] = \frac{1}{\prod_i(1-s_0\xi_0 u_i)(q;q)_{k-\ell}}$$

(the bracketed sum $\to [k-\ell]_q!$)

Note that $\tilde{F}_{\lambda}$ does not depend on s_0. Similarly, dual stable functions can be defined as:

$$\tilde{F}^{*}_{\lambda}(u_1,\dots,u_k) = F^{*}_{\lambda\cup 0^{k-\ell}}(u_1,\dots,u_k)\Big|_{s_0=0}.$$

The name stable comes from the property that

$$\tilde{F}_{\lambda}(u_1,\dots,u_{k-1},0) = \tilde{F}_{\lambda}(u_1,\dots,u_{k-1}).$$

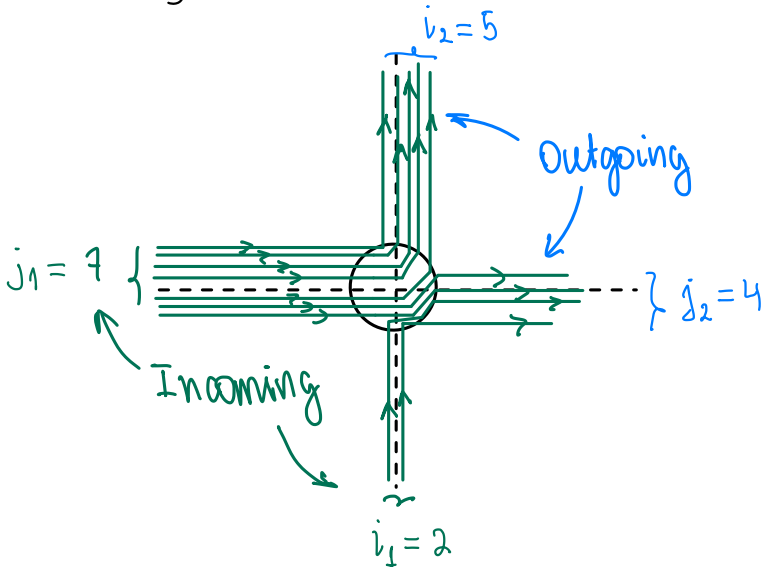
Stable spin Hall Littlewood functions satisfy a Cauchy-type identity:

$$\sum_{\lambda} \tilde{F}_{\lambda}(u_1,\dots,u_k)\ \tilde{F}^{*}_{\lambda}(v_1,\dots,v_m) = \prod_{i=1}^{n}\prod_{j=1}^{m} \frac{1-q u_i v_j}{1-u_i v_j},$$

where the sum runs over all partitions of arbitrary length.

2. The higher spin six vertex model

We admit a more general version of the usual six vertex model, where we admit each vertex can have at most I distinct incoming horizontal paths and at most J outgoing vertical paths, where $I, J \in \mathbb{Z}_{\geq 0}$. We impose the "arrow preservation property", where the number of incoming arrows equals the number of outgoing arrows. Note that this property fails at the boundaries; one way of working around this problem is to fix boundary conditions.



For the higher spin six vertex model, each vertex will be assigned a weight depending on two (complex) parameters q and s and a spectral parameter u.

Using the previous notation, the weights $w_{u,s}(i_1,i_2;j_1,j_2)$ of a given vertex must satisfy the arrow preservation rule $i_1+j_1=i_2+j_2$, i.e., so one can consider that occupied edges form up-right paths on the square grid.

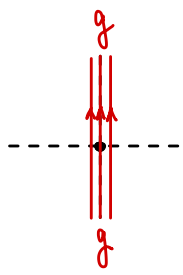
The explicit vertex weights are given by:

$$w_{u,s}(g,0;g,0) := \frac{1-sq^g u}{1-su}, \qquad w_{u,s}(g+1,0;g,1) := \frac{(1-s^2q^g)u}{1-su},$$

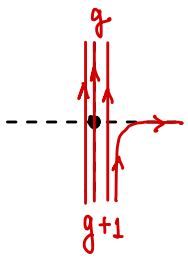
$$w_{u,s}(g,1;g,1) := \frac{u-sq^g}{1-su}, \qquad w_{u,s}(g,1;g+1,0) := \frac{1-q^{g+1}}{1-su},$$

where g is an arbitrary nonnegative integer. All other weights are assumed to be zero. Note that the weight of an empty vertex $(0,0;0,0)$ is equal to 1 and $w_{u,s}(i_1,i_2;j_1,j_2)$ is nonzero iff $j_1,j_2\in\{0,1\}$. Hence, for the higher spin six vertex model, $I=1$, while there is no limitation to J.

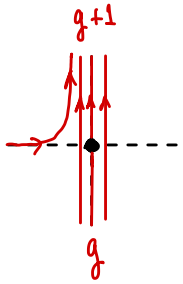
In general, the nontrivial weights fall into the following four categories:



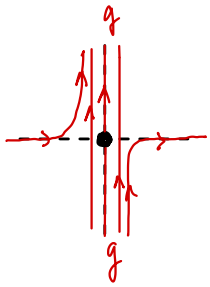
$$w_{u,s}(g,0;g,0) = \frac{1-suq^g}{1-su}$$



$$w_{u,s}(g+1,0;g,1) = \frac{(1-s^2q^g)u}{1-su}$$



$$w_{u,s}(g,1;g+1,0) = \frac{1-q^{g+1}}{1-su}$$

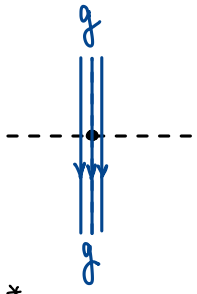


$$w_{u,s}(g,1;g,1) = \frac{u-sq^g}{1-su}$$

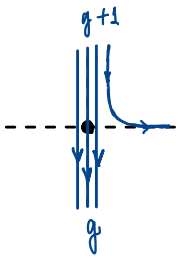
We also define dual weights $w^*_{u,v}$ for each vertex. The following formula expresses the relation between both weights.

$$W^*_{v,s}(i_1,j_1;i_2,j_2) = \frac{(s^2;q)_{i_1}(q;q)_{i_2}}{(s^2;q)_{i_2}(q;q)_{i_1}} W_{v,s}(i_2,j_1;i_1,j_2).$$

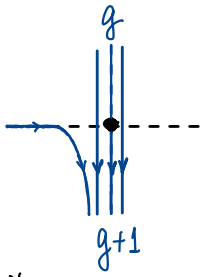
Note that the arrow preservation property now reads $i_2 + j_1 = i_1 + j_2$. Hence for the lattice with dual weights, we can think of occupied edges as forming down-right paths. The non trivial weights can be described by



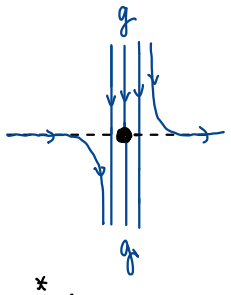
$$W^*_{v,s}(g,0;g,0) = \frac{1-svq^g}{1-sv}$$



$$W^*_{v,s}(g+1,0;g,1) = \frac{1-s^2q^g}{1-sv}$$



$$W^*_{v,s}(g,1;g+1,0) = \frac{(1-q^{g+1})v}{1-sv}$$



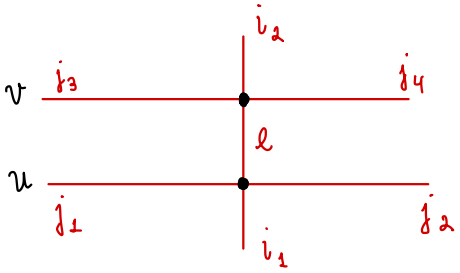
$$W^*_{v,s}(g,1;g,1) = \frac{v-sq^g}{1-sv}$$

Throughout this talk, we will use the following notation: the red model denotes up-right paths and the blue model denotes down-right paths.

2.1) The Red Yang-Baxter equation (RYBE)

The weights $W_{u,s}$ and $W^*_{v,s}$ are very special in that they verify a Yang-Baxter equation. We will deal with three cases of such equation.

To prove the basic properties of spin Hall-Littlewood symmetric functions, we will need a "red" version of YBE, i.e., a version that works well with the situation where all paths are up-right. For that, we define the vertex weights of a configuration of two vertices connected by a vertical edge.

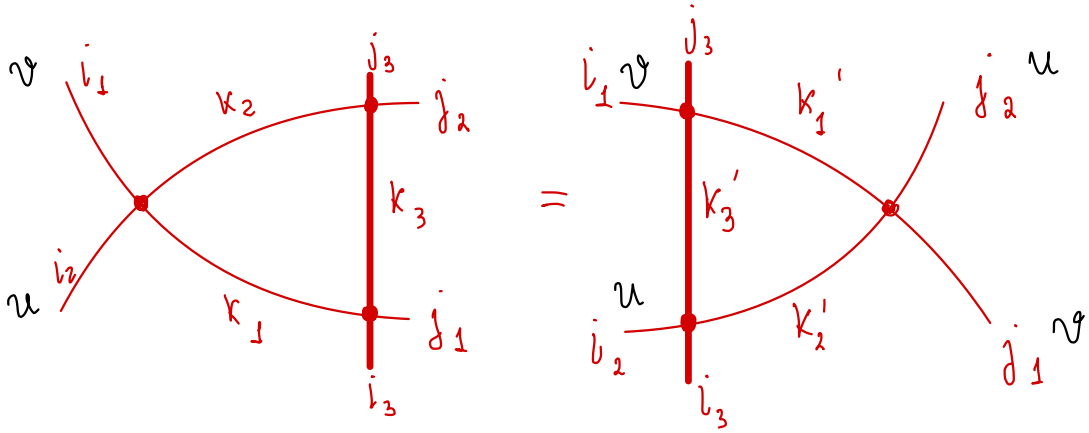


We define the two-vertex weight of such configuration, denoted by $W^{(i_1,i_2)}_{u,v;s}(j_1,j_3;j_2,j_4)$ as

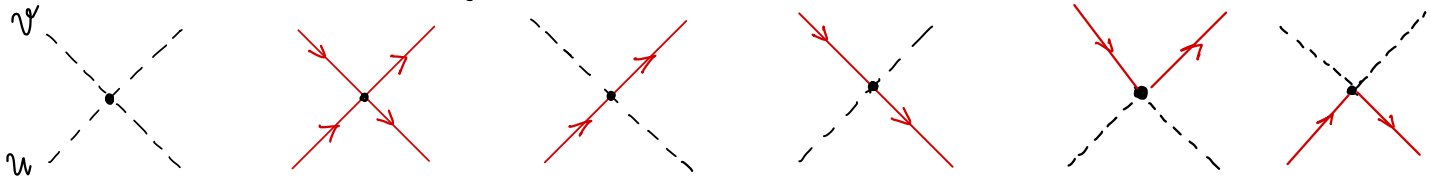
$$W^{(i_1,i_2)}_{u,v;s}(j_1,j_3;j_2,j_4) = \sum_{l\geq 0} W_{u,s}(i_1,j_1;l,j_2)\, W_{v,s}(l,j_3;i_2,j_4), \quad j_1,j_2,j_3,j_4 \in \{0,1\}.$$

The number $W^{(i_1,i_2)}_{u,v;s}(j_1,j_3;j_2,j_4)$ represents the weight of a two vertex configuration with the number of incoming and outcoming arrows j_1, j_3, j_2, j_4 fixed. The number $l\geq 0$ of arrows along the inside edge is arbitrary, but due to arrow preservation, no more than one value of l can contribute to the sum.

We also need to define the weights on red cross vertex which also depend on q (but not on the spectral parameters).



Note that along "red-red" cross vertices like $(i_2, i_1; k_2, k_1)$ above, arrow preservation implies that $i_1, i_2, k_1, k_2 \in \{0,1\}$, which is the defining condition of the classic six vertex model. We then define the following vertex weights, denoted by $R_{u,v}(i_2,i_1;k_2,k_1)$.



Given those vertex weights, the picture above can be rewritten as

$$\sum_{k_1,k_2,k_3}' R_{u,v}(i_2,i_1;k_2,k_1)\, W_{u,s}(k_3,k_2;j_3,j_2)\, W_{v,s}(i_3,k_1;k_3,j_1)$$

$$= \sum_{k_3',k_2',k_3'} W_{v,s}(k_3',i_1;j_3,k_1')\, W_{u,s}(i_3,i_2;k_3',k_2')\, R_{u,v}(k_2',k_1';j_2,j_1)$$

which we will call the Red Yang-Baxter Equation (RYBE).

2.2) Spin Hall-Littlewood symmetric functions as partition functions

Fix q and the parameters s_x, ξ_x, where $x = 0, 1, 2, \ldots$ is the horizontal coordinate. We define the spin Hall-Littlewood function $F_\lambda(u_1, \ldots, u_N)$, $\lambda \in \text{Sign}_N$ as the partition function of up-right paths in $\mathbb{Z}_{\geq 0} \times \{1, \ldots, N\}$, with the following restrictions:

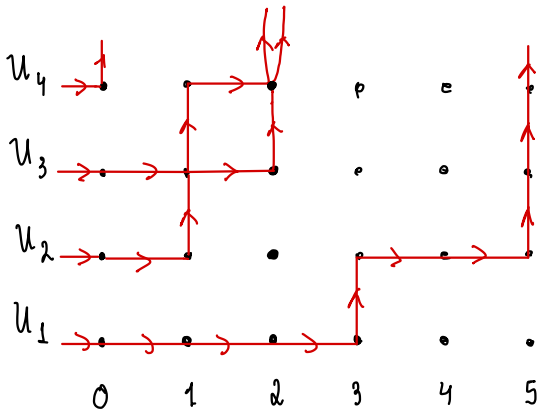
a) A path enters the region at each location $(0,i)$, $i = 1, \ldots, N$, on the left boundary;

b) A path exits the region at locations (λ_j, N), $j = 1, \ldots, N$ on the top boundary;

c) No paths enter through the bottom boundary or escape to infinity on the far right.

d) At each vertex (x,i), $x \in \mathbb{Z}_{\geq 0}$, the vertex weight is $w_{u_i \xi_x, s_x}$.

Ex.: $\lambda = (5,2,2,0) \in \text{Sign}_4$.

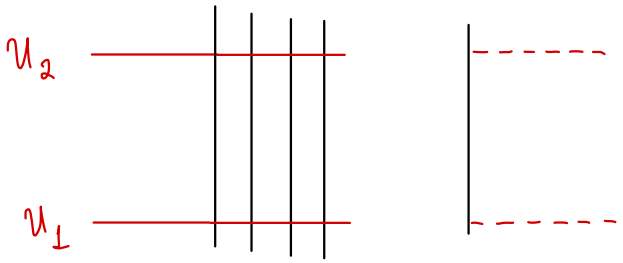


An example of configuration that contributes to the calculation of the partition function $F_\lambda(u_1,u_2,u_3,u_4) =$

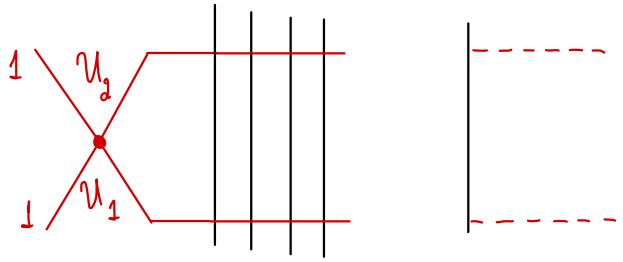
$$= \sum_{\lambda} \prod_{i=1}^{4} \prod_{x=0}^{\lambda_i} w_{u_i \varepsilon_{ix}, s_x}.$$

Theorem 1: The partition function $F_\lambda(u_1,\dots,u_N)$ is symmetric w.r.t. the u_i's.

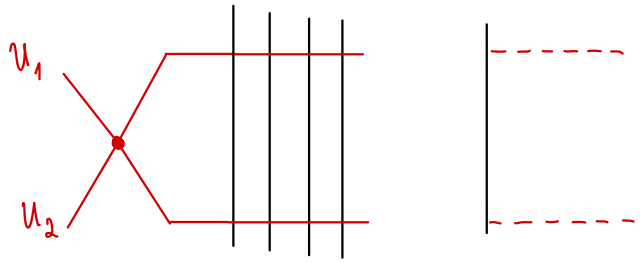
Proof: By induction, it is enough to consider the case of two (adjacent) lines. Consider the situation where the following picture holds:



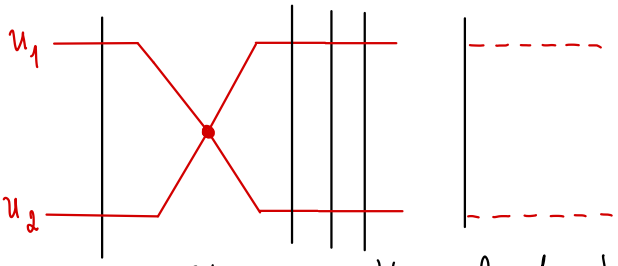
By the boundary conditions, the two edges at the end have to be dotted. The black lines may have an arbitrary number of arrows. The partition function of this model is $F_\lambda(u_1,u_2)$. We introduce a crossing of the two red lines.



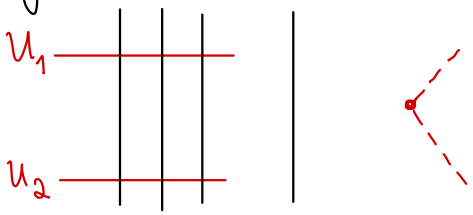
The introduction of a new vertex should affect the partition function F_λ, but that change is trivial: the only nonzero crossing of two red edges is  , whose weight is 1. Hence, the introduction of the crossing does not affect the terms of the formula.



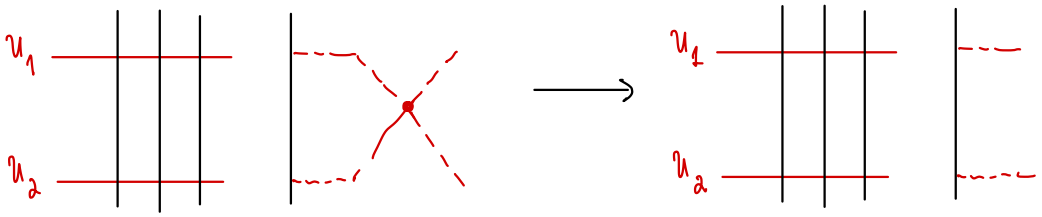
We now use the RYBE successively to push the crossing through the black lines.



When the crossing goes through the last black line, again by the boundary conditions, no solid lines should be able to scape to infinity. Thus, the last crossing should look like this



Again, the only nonzero crossing in this case is  , whose weight is 1. This extra vertex can be removed without changing the partition function.



, which is a model whose partitition function is $F_\lambda(u_2, u_1)$. Hence $F_\lambda(u_1, u_2) = F_\lambda(u_2, u_1)$.

□

Theorem 2: Let $\varphi_k(u) := \frac{1-q}{1-s_k\xi_k u} \prod_{j=0}^{k-1} \frac{\xi_j u - s_j}{1-s_j\xi_j u}$, $k \geq 0$.

Then

$$F_\lambda(u_1,\dots,u_N) = \sum_{\sigma\in S_N} \sigma\left(\prod_{i<j} \frac{u_i - q u_j}{u_i - u_j} \prod_{i=1}^{N} \varphi_{\lambda_i}(u_i) \right).$$

The proof will be shown later (last page)

2.3) The Blue and Red-Blue Yang-Baxter Equations

First, we define the dual spin Hall-Littlewood symmetric functions F^*_λ in a similar fashion to the F_λ: the functions $F^*_\lambda(u_1,\dots,u_N;q)$ are the partition functions of down-right paths in $\mathbb{Z}_{\geq 0}\times\{1,\dots,N\}$, with the following restrictions:

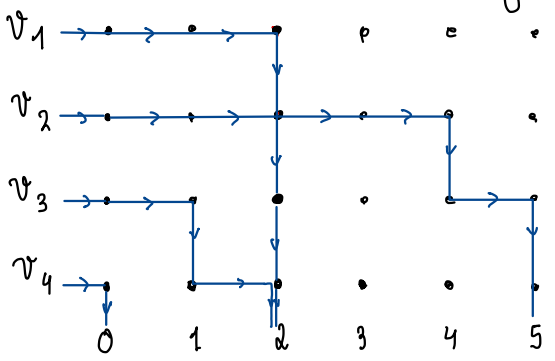
a) A path enters the region at each location $(0,i)$, $i=1,\dots,N$, on the left boundary;

b) A path exits the region at locations $(\lambda_j,0)$, $j=1,\dots,N$ on the bottom boundary;

c) No paths enter through the top boundary or escape to infinity on the far right.

d) At each vertex (x,i), $x\in\mathbb{Z}_{\geq 0}$, the vertex weight is $w^*_{u_i\xi_i^{-1},s_x}$.

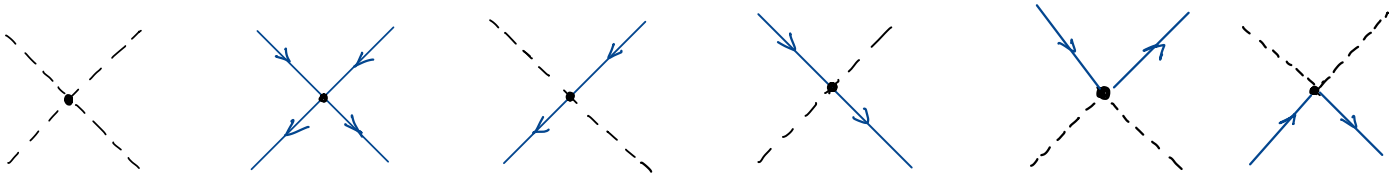
Ex.: $\lambda = (5,2,2,0) \in \mathrm{Sign}_4$.



An example of a config. that contributes to the calculation of the partition function $F^*_\lambda(v_1, v_2, v_3, v_4; q)$

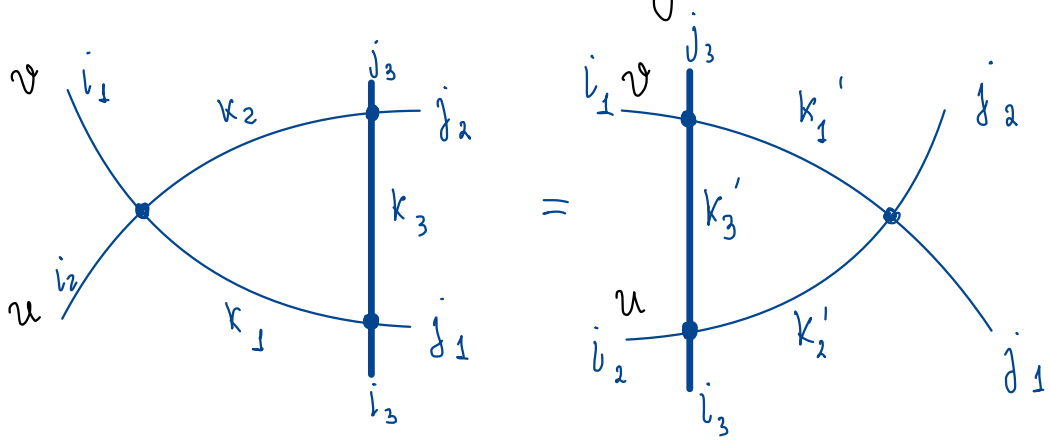
$$= \sum_{\lambda} \prod_{i=1}^{4} \prod_{x=0}^{\lambda_i} w_{u_i \xi_x^{-1}, s_x}.$$

To construct a blue version of the young-Baxter equation, we need the blue cross weights.



1 $\quad$ 1 $\quad$ $\frac{q(u-v)}{u-qv}$ $\quad$ $\frac{u-v}{u-qv}$ $\quad$ $\frac{(1-q)v}{u-qv}$ $\quad$ $\frac{(1-q)u}{u-qv}$

We now construct the following picture.



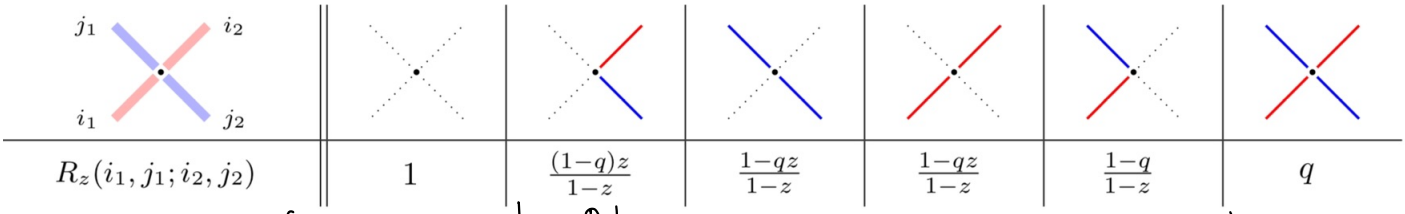
And we get the Blue yang-Baxter Equation (BYBE). This formula can be derived as a direct consequence of the RYBE and the formula for dual weights. In fact, this is the reason why the formula for blue cross weights is the same.

$$\sum_{k_1,k_2,k_3} R_{u,v}(i_2,i_1;k_2,k_1)\, W^{*}_{u,s}(k_3,k_2;j_3,j_2)\, W^{*}_{v,s}(i_3,k_1;k_3,j_1)$$

$$= \sum_{k_3',k_2',k_3'} W^{*}_{v,s}(k_3',i_1;j_3,k_1')\, W^{*}_{u,s}(i_3,i_2;k_3',k_2')\, R_{u,v}(k_2',k_1';j_2,j_1)$$

(BYBE)

Finally, to write down the most general case, we define red-blue cross vertex weights R_z which depend on q (but not on s).



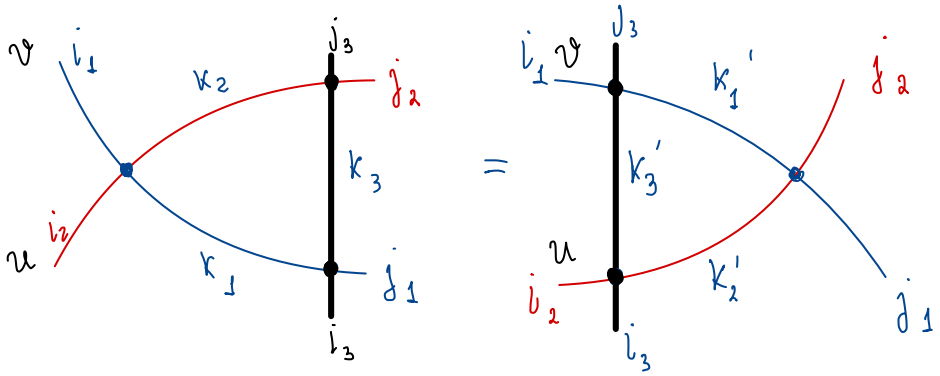
$R_z(i_1,j_1;i_2,j_2)$	1	$\frac{(1-q)z}{1-z}$	$\frac{1-qz}{1-z}$	$\frac{1-qz}{1-z}$	$\frac{1-q}{1-z}$	q

Theorem 3 (The Red-Blue Yang-Baxter Equation).

For $i_1,i_2,j_1,j_2 \in \{0,1\}$ and $i_3,j_3 \in \mathbb{Z}_{\geq 0}$, we have

$$\sum_{k_1,k_2,k_3} R_{u,v}(i_2,i_1;k_2,k_1)\, W^{*}_{v,s}(i_3,k_1;k_3,j_1)\, W_{u,s}(k_3,k_2;j_3,j_2) =$$

$$= \sum_{k_1',k_2',k_3'} W^{*}_{v,s}(k_3',i_1;j_3,k_1')\, W_{u,s}(i_3,i_2;k_3',k_2')\, R_{u,v}(k_2',k_1';j_2,j_1).$$



Note that the cross vertex weights on the YBE do not depend on s, and depend on u,v only through their product uv. Hence, the YBE remains unchanged for a whole family of weights $(W_{u\xi,s};\, W^{*}_{v/\xi,s})$ for arbitrary ξ and s.

3) Proof of the Cauchy-type identity for stable functions

Our goal is to prove the following identity.

$$\sum_{\lambda} \tilde{F}_{\lambda}(u_1,\dots,u_N)\,\tilde{F}^{*}_{\lambda}(v_1,\dots,v_M) = \prod_{i=1}^{N}\prod_{j=1}^{M}\frac{1-q u_i v_j}{1-u_i v_j},$$

where λ runs over all partitions.

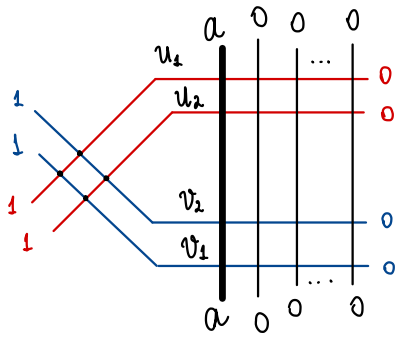
First, we have to make sure such sum converges. To that effect, we assume the spectral parameters u_i, v_j and the parameters s_x, ξ_x satisfy the condition

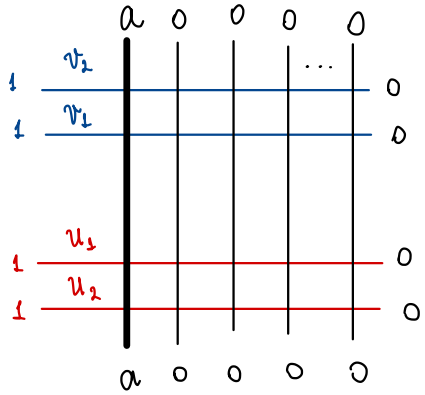
$$(*)\qquad \left|\frac{\xi_x u_i - s_x}{1-s_x\xi_x u_i}\cdot\frac{v_j/\xi_x - s_x}{1-s_x v_j/\xi_x}\right| \leqslant 1-\varepsilon < 1,$$

for all i,j and all $x = 0, 1, 2, \dots$.

Next, we look at two lattice configurations combining red and blue edges, where the zeroth vertical edge has a arrows, with $a \in \{0,1,2,\dots\} \cup \{+\infty\}$. The other boundary condition are shown in the pictures. Let $r := q^a$. Assume, initially, that $N = M$.

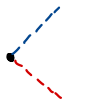
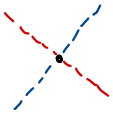
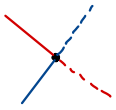
For $N=2$

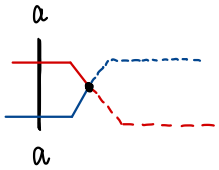


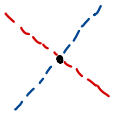


Let's denote the partition function of the model to the left by Z^{r}_{N} and the partition function of the model to the right by S^{r}_{N}. Note that these functions depend on the extra parameters s_0, ξ_0 and $r = q^a$. If $a = \infty$, we define $r = 0$

Theorem 4: If u_i, v_j satisfy condition (*), then $Z_N^\gamma = S_N^\gamma$.

We start with the configuration for Z_N^γ. We use the red-blue Yang-Baxter equation to move all cross vertices to the right; by the boundary conditions, we know the traversed cross vertices will have  . Thus, the only two nonzero possibilities are  or 

If we end up with 

, these cross vertices will introduce infinitely many factors of the form $W_{u_i \xi_x, s_x}(0,1;0,1) W^*_{v_j/\xi_x, s_x}(0,1;0,1)$, but by condition (*) these factors are smaller than $1-\varepsilon$ in absolute value, so they don't contribute to the partition function. The remaining cross vertices  have weight 1, so they don't contribute to the partition function and can be eliminated.

Hence, we arrive at the configuration for S_N^γ. □

Theorem 5: Let u_i, v_j satisfy (*). When $\gamma \neq 0$, then

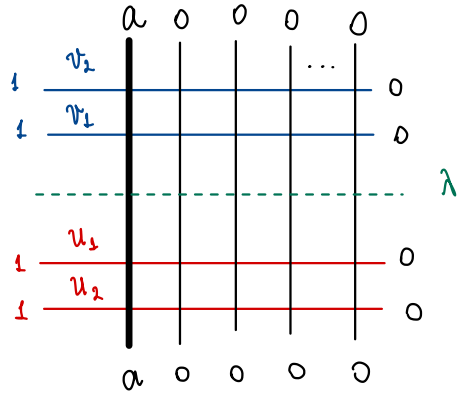
$$S_N^\gamma(u_1,\dots,u_N; v_1,\dots,v_N; \gamma^{-1}s_0) = \prod_{j=1}^N \frac{1-\xi_0 s_0 u_j}{1-s_0\gamma^{-1}u_j} \cdot \frac{1-s_0 v_j/\xi_0}{1-s_0\gamma^{-1}v_j/\xi_0} \times$$
$$\times \sum_{\lambda \in \mathrm{Sign}_N} \frac{(\gamma q;q)_{N-\ell(\lambda)}}{(q;q)_{N-\ell(\lambda)}} \cdot \frac{(\gamma^{-1}s_0^2;q)_{N-\ell(\lambda)}}{(s_0^2;q)_{N-\ell(\lambda)}} F_\lambda(u_1,\dots,u_N) F^*_\lambda(v_1,\dots,v_N).$$

where $\ell(\lambda)$ = # of nonzero parts of λ.

For $\gamma = 0$,

$$S_N^{\gamma=0}(u_1,\dots,u_N; v_1,\dots,v_N; s_0) = \prod_{j=1}^N \frac{1}{(1-s_0\xi_0 u_j)(1-s_0 v_j/\xi_0)} \sum_{\mu \subseteq (N)^N} \tilde{F}_\mu(u_1,\dots,u_N)\, \tilde{F}^*_\mu(v_1,\dots,v_N).$$

Proof: We assume $a \geq N$. Fix an arbitrary path configuration contributing to the partition function S_N^{γ}. Let $\lambda \in \text{Sign}_N$ encode the intermediate arrow configurations between the red and blue parts.



Look at the red part first. Apart from the zeroth column, the weights $w_{u_j\xi_x, s_x}$, $x \geq 1$ are the same as in the definition of $F_\lambda(u_1, \dots, u_N)$. The number of arrows leaving the red half into the blue half is $a + N - \ell(\lambda)$. In the zeroth column, there are two types of vertices:

$$w_{u_j\xi_0, s_0/\gamma}(a+i, 1; a+i, 1) = \frac{u_j\xi_0 - s_0/\gamma q^{a+i}}{1 - s_0/\gamma u_j\xi_0} = \frac{u_j\xi_0 - s_0 q^i}{1 - s_0\xi_0 u_i/\gamma}$$

$$w_{u_j\xi_0, s_0/\gamma}(a+i, 1; a+i+1, 0) = \frac{1 - \gamma q^{i+1}}{1 - s_0 u_j \xi_0/\gamma}$$

The number of vertices of the second type is $N - \ell(\lambda)$, since that is the quantity of paths that leave the zeroth column. Such vertices correspond to $0 \leq i \leq N - \ell(\lambda) - 1$. To remove the dependency on γ in the numerator of the second type, we should factor out $\prod_{i=0}^{N-\ell(\lambda)-1}(1 - \gamma q^{i+1})$. We now take care of the dependency

$$\frac{\prod_{i=0}^{N-\ell(\lambda)-1}(1 - \gamma q^{i+1})}{\prod_{i=0}^{N-\ell(\lambda)-1}(1 - q^{i+1})} = \frac{(q\gamma, \gamma)_{N-\ell(\lambda)}}{(q;q)_{N-\ell(\lambda)}}.$$

on r at the denominators by factoring out $\prod_{j=1}^{N} \frac{1-\xi_0 s_0 u_j}{1-\xi_0 s_0 r^{-1} u_j}$.
After such changes, the weights along the zeroth column match the ones in $F_\lambda(u_1,\dots,u_N)$. Hence, the partition function for the red part is

$$\frac{(qr;r)_{N-\ell(\lambda)}}{(q;q)_{N-\ell(\lambda)}} \prod_{j=1}^{N} \frac{1-\xi_0 s_0 u_j}{1-\xi_0 s_0 r^{-1} u_j} \cdot F_\lambda(u_1,\dots,u_N).$$

The calculation for the blue half is extremely similar. In the zeroth column, we also have two types of vertices.

$$w^*_{v_j/\xi_0, s_0/r}(a+i,1;a+i,1) = \frac{v_j/\xi_0 - s_0 q^i}{1-s_0 r^{-1} v_j/\xi_0}$$

$$w^*_{v_j/\xi_0, s_0/r}(a+i+1,1;a+i,0) = \frac{1-s_0^2 r^{-1} q^i}{1-s_0 r^{-1} v_j/\xi_0}.$$

We now take a common factor of $\frac{(s_0^2 r^{-1};q)_{N-\ell(\lambda)}}{(s_0^2;q)_{N-\ell(\lambda)}} \prod_{j=1}^{N} \frac{(1-s_0 v_j/\xi_0)}{(1-s_0 r^{-1} v_j/\xi_0)}$

to have the weights of the zeroth column coincide with those in $F^*_\lambda(v_1,\dots,v_N)$. Putting all together and summing over λ, we have

$$S^r_N(u_1,\dots,u_N;v_1,\dots,v_N;r^{-1}s_0) = \prod_{j=1}^{N} \frac{1-\xi_0 s_0 u_j}{1-s_0 r^{-1} u_j} \cdot \frac{1-s_0 v_j/\xi_0}{1-s_0 r^{-1} v_j/\xi_0} \times$$

$$\times \sum_{\lambda \in \mathrm{Sign}_N} \frac{(rq;q)_{N-\ell(\lambda)}}{(q;q)_{N-\ell(\lambda)}} \cdot \frac{(r^{-1}s_0^2;q)_{N-\ell(\lambda)}}{(s_0^2;q)_{N-\ell(\lambda)}} F_\lambda(u_1,\dots,u_N) F^*_\lambda(v_1,\dots,v_N).$$

If $a \to \infty$, $r \to 0$, replace $r^{-1}s_0 \to s_0$, and the pre factor becomes $\prod_{j=1}^{N} \frac{1}{(1-\xi_0 s_0 u_j)(1-s_0 v_j/\xi_0)}$, while the sum becomes

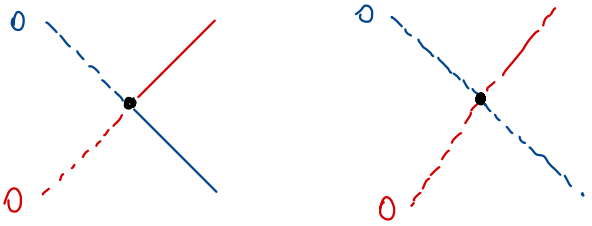
$$\sum_{\substack{\mu \\ \ell(\mu) \le N}} \tilde{F}_\lambda(u_1,\dots,u_N) \tilde{F}^*_\lambda(v_1,\dots,v_N), \text{ when } s_0 = 0.$$

Theorem 6:

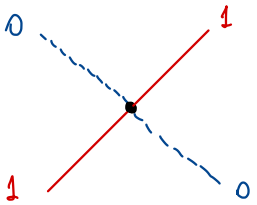
$$Z_N^{r=0}(u_1,\dots,u_N;v_1,\dots,v_N;s_0) = \prod_{j=1}^{N} \frac{1}{(1-s_0\xi_0 u_j)(1-s_0 v_j/\xi_0)} \prod_{i,j=1}^{N} \frac{1-q\,u_i v_j}{1-u_i v_j}.$$

Proof: For the case $a=\infty$, note that the weights $w_{u_j;\xi_0,s_0}$ and $w^*_{v_j/\xi_0,s_0}$ do not depend on the number of incoming horiz. arrows from the left, and are equal to $\frac{1}{1-s_0 u_j \xi_0}$ and $\frac{1}{1-s_0 v_j/\xi_0}$, respectively.

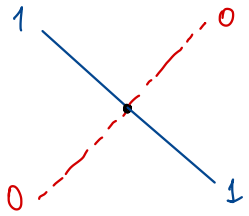
For the red-blue cross vertices, we note that



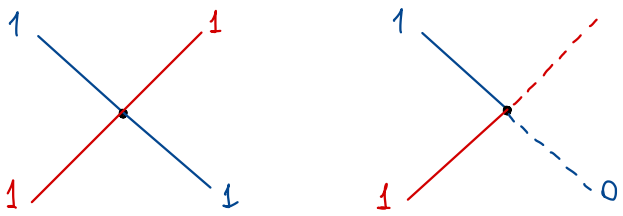
$$\frac{(1-q)uv}{1-uv} + 1 = \frac{1-quv}{1-uv}$$



$$\frac{1-quv}{1-uv}$$



$$\frac{1-quv}{1-uv}$$



$$q + \frac{1-q}{1-uv} = \frac{1-quv}{1-uv}$$

and thus $\sum_{i_2,j_2} R_{uv}(i_1,j_1;i_2,j_2) = \frac{1-quv}{1-uv}$ for all i_1,j_1.

Hence, the partition function factors, and its formula is the one given above. □

Cauchy-type identity for stable spin Hall-Littlewood symmetric functions.

From the equality $S_N^{r=0} = Z_N^{r=0}$, we have

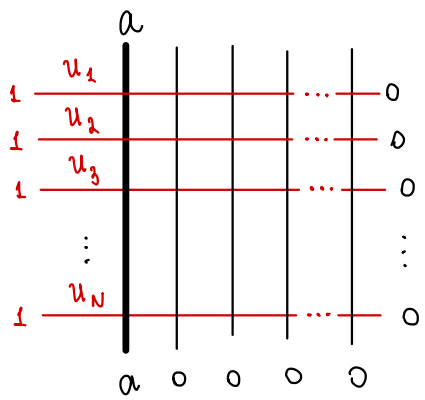
$$\sum_{\substack{\mu \\ \ell(\mu)\le N}} \tilde{F}_\mu(u_1,\dots,u_N)\,\tilde{F}^*_\mu(v_1,\dots,v_N) = \prod_{i,j=1}^{N} \frac{1-q\,u_i v_j}{1-u_i v_j},$$

as desired.

Proof of Theorem 4: The proof is by induction on N.

First, let's denote by $F_\lambda^{[a]}$ the partition function of the red portion of the lattice shown in the proof of Thorem 5, with $a \ge N$. Note that $F_\lambda^{[0]} = F_\lambda$.

By symmetry of F_λ, we can reorder the lattice to look like this:



λ adds one solid line above the point (λ_i, i).

One more consideration to make. If $\lambda_N > 0$, then we can consider the partition $\hat{\lambda}$, where $\hat{\lambda}_j = \lambda_j - \lambda_N$, with relabelled parameters $\hat{\xi}_x = \xi_{x+\lambda_N}$, $\hat{s}_x = s_{x+\lambda_N}$. Then,

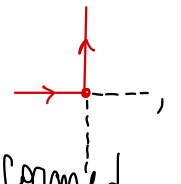
$$F_\lambda(u_1,\dots,u_N) = \prod_{i=1}^{N}\prod_{j=0}^{\lambda_N-1} \frac{\xi_j u_i - s_j}{1-s_j\xi_j u_i}\, F_{\hat{\lambda}}(u_1,\dots,u_N;\hat{\xi}_x;\hat{s}_x),$$

so it is enough to consider the case where $\lambda_N = 0$.

Now, we think of F_λ as a rational function of ξ_0.

At $x=0$, weights are either $\dfrac{u_i\xi_0 - s_0 q^g}{1-\xi_0 u_i s_0}$ or $\dfrac{1-q^{g+1}}{1-\xi_0 u_i s_0}$, so $F_\lambda \to 0$ as $\xi_0 \to 0$.

Thus, one can rewrite $F_\lambda(\xi_0) = \dfrac{G(\xi_0)}{\prod_{i=1}^{N}(1-\xi_0 u_i s_0)}$, where G is a polynomial such that $\deg G \leq N-1$. Now, F_λ is completely determined by the zeros of $G(\xi_0)$. There is a finite number of zeros, at the points where $u_i\xi_0 - s_0 = 0$, that is, at $\xi_0 = s_0 u_i^{-1}$. By symmetry again, it is enough to consider the case $i=N$.

Since $\lambda_N = 0$, the lines at $(0,1)$ look like  , whose weight is $\dfrac{1-q}{1-s_0 u_N \xi_0}$. We move to the sublattice formed by excluding the last line, indexed by u_N. Let $\tilde{\lambda} = (\lambda_1, \dots, \lambda_{N-1})$

Note that $F_\lambda(\xi_0)\Big|_{\xi_0 = u_N^{-1}s_0} = F_\lambda(u_1,\dots,u_N)\Big|_{\xi_0 = u_N^{-1}s_0} = \dfrac{1-q}{1-s_0 u_N \xi_0} F^{[1]}_{\tilde{\lambda}}(u_1,\dots,u_{N-1}; s_0 q^{-1}, s_1, \dots, s_{\lambda_1}; s_0 u_N^{-1}, \xi_1, \dots, \xi_{\lambda_1}) =$

$= \dfrac{1-q}{1-s_0^2} F^{[1]}_{\tilde{\lambda}}(u_1,\dots,u_{N-1}; s_0 q^{-1}, s_1, \dots, s_{\lambda_1}; s_0 u_N^{-1}, \xi_1, \dots, \xi_{\lambda_1})$

From the result of Theorem 5 with $r = q^1$, we have

$$F^{[1]}_{\tilde{\lambda}}(u_1,\dots,u_{N-1}; s_0 q^{-1}, s_1, \dots, s_{\lambda_1}; s_0 u_N^{-1}, \xi_1, \dots, \xi_{\lambda_1}) = \frac{(q^2;q)_{N-\ell(\lambda)}}{(q;q)_{N-\ell(\lambda)}} \prod_{j=1}^{N} \frac{1-\xi_0 s_0 u_j}{1-\xi_0 s_0 q^{-1} u_j} \cdot F_{\tilde{\lambda}}(u_1,\dots,u_{N-1}; s_0 q^{-1}, s_1, \dots, s_{\lambda_1}; s_0 u_N^{-1}, \xi_1, \dots, \xi_{\lambda_1})$$

With the substitution $s_0 \to s_0 q$, we find

$$F^{[1]}_{\tilde{\lambda}}(u_1,\dots,u_{N-1}; s_0 q^{-1}, s_1, \dots, s_{\lambda_1}; s_0 u_N^{-1}, \xi_1, \dots, \xi_{\lambda_1}) = \frac{(q^2;q)_{\overbrace{N-\ell(\lambda)}^{m_0}}}{(q;q)_{N-\ell(\lambda)}} \prod_{j=1}^{N} \frac{1-\xi_0 s_0 u_j q}{1-\xi_0 s_0 u_j} \cdot F_{\tilde{\lambda}}(u_1,\dots,u_{N-1}; s_0, s_1, \dots, s_{\lambda_1}; s_0 u_N^{-1}, \xi_1, \dots, \xi_{\lambda_1})$$

$$= [m_0]_q \cdot \prod_{j=1}^{N} \frac{1-\xi_0 s_0 u_j q}{1-\xi_0 s_0 u_j} \cdot F_{\tilde{\lambda}}(u_1,\dots,u_{N-1}; s_0, s_1, \dots, s_{\lambda_1}; s_0 u_N^{-1}, \xi_1, \dots, \xi_{\lambda_1})$$

Finally, we can write the following recursion for F_λ:

$$F_\lambda(u_1,\dots,u_N)\Big|_{\xi_0 = u_N^{-1}s_0} = \frac{1-q}{1-s_0^2}\cdot[m_0]_q \cdot \prod_{j=1}^{N} \frac{1-\xi_0 s_0 u_j q}{1-\xi_0 s_0 u_j}\cdot F_{\tilde{\lambda}}(u_1,\dots,u_{N-1};\, s_0, s_1,\dots,s_{\lambda_1};\, s_0 u_N^{-1}, \xi_{1_1},\dots,\xi_{\lambda_1})$$

It can be shown, with a certain deal of work, that the symmetrization formula

$$F_\lambda(u_1,\dots,u_N) = \sum_{\sigma\in S_N} \sigma\left(\prod_{i<j}\frac{u_i - q u_j}{u_i - u_j}\prod_{i=1}^{N}\varphi_{\lambda_i}(u_i)\right)$$

also satisfies the same recursion as a function of ξ_0. For $N=1$, both formulas for F_λ coincide and are equal to $F_\lambda(u_1) = \frac{1-q}{1-s_{\lambda_1}\xi_{\lambda_1}u_1}$. The result now follows from induction and the above recursion.

□